
DISSOCIATING PERSEVERATIVE RESPONDING FROM LEARNING-RATE EFFECTS IN TRAUMA: A COMPUTATIONAL MODELING APPROACH

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ABSTRACT

Trauma exposure alters reinforcement learning (RL) and working memory (WM) processes, but the computational mechanisms remain unclear. We fitted seven computational models, spanning classical Q-learning (M1), WM-RL hybrids with optional perseveration kernels (M2, M3, M5, M6a, M6b), and a joint choice-plus-response-time model using a Linear Ballistic Accumulator (M4), to data from 162 participants completing a reinforcement learning working memory task. The preferred model by both AIC and BIC was M6b, which decomposes perseverative responding into choice-level and stimulus-level components via a stick-breaking share parameter (κ_{share}). Parameter-recovery simulations on synthetic data show that the two perseveration parameters (κ_{total} , κ_{share}) are identifiable ($r \geq 0.93$), but the base RLWM parameters (learning rates, WM decay, WM reliability, capacity, attentional noise) fall short of the $r \geq 0.80$ criterion. Trauma-parameter associations were marginal: the strongest effects were a positive association between the perseveration kernel (κ_{total} , M6b; κ , M3) and lifetime trauma exposure (LEC-5 Total Events), which survived false-discovery-rate correction only in M3. We report all associations alongside uncorrected, FDR-BH, and Bonferroni-corrected p-values and situate the findings as hypothesis-generating, pending confirmation in an independent sample.

Keywords reinforcement learning · working memory · trauma · perseveration · computational psychiatry · PTSD

1 Introduction

Trauma exposure produces lasting changes in learning and decision-making that extend well beyond the initial stressor (??). Computational psychiatry has begun to characterize these changes in terms of reinforcement learning (RL) parameters: altered learning rates, impaired value representations, and disrupted prediction-error signals (???). Yet a fundamental ambiguity remains: apparent learning-rate deficits following trauma may reflect not genuine insensitivity to outcomes, but rather motor perseveration — a tendency to repeat prior actions independent of their value.

Working memory (WM) contributes substantially to behavior on RL tasks, particularly at low set sizes where stimuli fit within WM capacity (??). The WM-RL hybrid model of ? formalizes this interaction and introduces a perseveration parameter (κ) that captures action repetition tendencies separable from outcome-based learning. By applying this framework to a trauma-exposed sample, we can ask whether post-reversal failures following trauma reflect elevated κ (motor habit), reduced α (outcome insensitivity), or both.

This paper describes the fitting of seven hierarchically nested computational models (M1 Q-learning through M6b dual-perseveration) to behavioral data from 162 participants stratified by trauma history and post-traumatic stress symptom severity. We compare models by AIC/BIC, identify the winning model, and examine how its parameters relate to trauma group membership and continuous symptom measures from the IES-R (?) and LEC-5 (?).

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2 Methods

2.1 Participants

Participants were included in all downstream analyses only if they met the v4.0 canonical inclusion criteria implemented in `config.get_analysis_cohort()`, which enforces the intersection of three gates: (1) **task completeness** — at least 400 trials and at least eight main-task blocks; (2) a **performance check** — overall accuracy above 0.40 AND mean accuracy over the last 5 main-task blocks above 0.50 (the late-block criterion confirms the participant learned rather than performing at or near chance throughout the session, since chance = $1/3 \approx 0.333$); (3) **scale completeness** — non-missing values on every required survey column (LEC-5 total events, IES-R total, IES-R intrusion, IES-R avoidance, IES-R hyperarousal). The resulting analysis cohort comprised **162** participants. Trauma exposure and symptom severity were assessed via the Life Events Checklist for DSM-5 (LEC-5; ?) and the Impact of Event Scale–Revised (IES-R; ?). Participants were classified into two groups based on LEC-5 endorsement and IES-R total score: **Trauma - No Ongoing Impact** ($n = 63$) and **Trauma - Ongoing Impact** ($n = 114$).

Scale distributions and pairwise correlations are summarized in Figure ??.

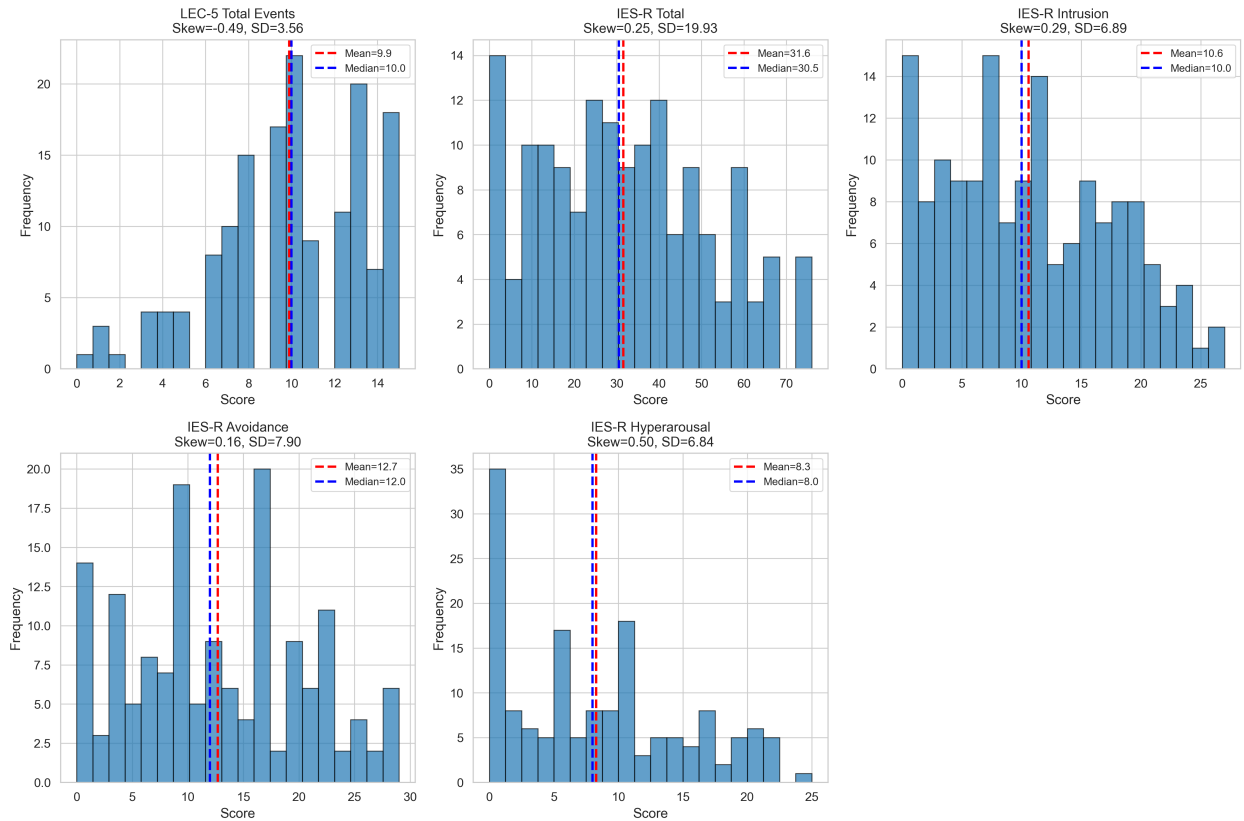


Figure 1: Distributions and pairwise correlations of the trauma predictors. **Left:** histograms of LEC-5 total events and IES-R total and subscale scores. **Right:** Pearson correlation matrix. LEC total correlates only weakly with IES total ($r = 0.13$), justifying inclusion of both predictors. IES subscales correlate strongly with IES total ($r = 0.91$ – 0.93), requiring the Gram–Schmidt residualization described in Model Fitting. Figure source: manuscript/figures/scale_distributions.png.

2.2 Task

Participants completed the RLWM task from ? implemented in jsPsych. On each trial, a stimulus was presented and participants selected one of three response options (J, K, L) using the keyboard. Feedback (correct/incorrect) was provided after each response. Stimuli were organized into blocks with set sizes of 2, 3, 5, or 6 stimuli (set size 4 excluded), with rare reversals of correct response mappings occurring after 12–18 consecutive correct responses. The task comprised two practice blocks (static and dynamic) followed by up to 21 main-task blocks (see Section ?? for inclusion thresholds).

2.3 Computational Models

We fitted seven models of increasing complexity to each participant’s trial-by-trial choices (see Section ?? for complete parameter listings). All WM-RL models share a core architecture in which behaviour arises from a mixture of reinforcement learning (RL) and working memory (WM) policies, with the WM contribution weighted by capacity and set size.

Core WM-RL architecture. On each trial, WM decays globally toward a uniform baseline, then overwrites the observed stimulus–action–reward association:

$$\text{WM}(s, a) \leftarrow (1 - \phi) \text{WM}(s, a) + \phi/n_A \quad (\text{decay}), \quad \text{WM}(s, a) \leftarrow r \quad (\text{update})$$

RL values are updated via an asymmetric delta rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \delta, \quad \delta = r - Q(s, a)$$

Both systems produce softmax choice probabilities ($\beta = 50$, fixed), combined via an adaptive weight:

$$\omega = \rho \cdot \min(1, K/n_s), \quad p_{\text{hybrid}}(a | s) = \omega p_{\text{WM}} + (1 - \omega) p_{\text{RL}}$$

Epsilon noise and dual perseveration (M6b). The winning model adds uniform noise and two perseveration kernels (global and stimulus-specific) via a stick-breaking parameterization:

$$\kappa = \kappa_{\text{total}} \cdot \kappa_{\text{share}}, \quad \kappa_s = \kappa_{\text{total}} \cdot (1 - \kappa_{\text{share}})$$

The final choice probability is:

$$p(a | s) = (1 - \kappa - \kappa_s) \left[\frac{\varepsilon}{n_A} + (1 - \varepsilon) p_{\text{hybrid}} \right] + \kappa \mathbf{1}[a = a_{\text{last}}] + \kappa_s \mathbf{1}[a = a_{\text{last}}^{(s)}]$$

where a_{last} is the globally previous action and $a_{\text{last}}^{(s)}$ is the previous action for stimulus s .

Model Parameter Glossary

Learning rate. α governs the speed of Q-value updates following prediction errors. A single symmetric learning rate is used for both positive and negative prediction errors.

WM decay (ϕ). Controls how rapidly WM representations degrade between trials. Higher ϕ means faster information loss from working memory, forcing greater reliance on the slower RL system.

WM reliability (ρ). Scales the overall contribution of WM to the action policy. Lower ρ means the participant relies less on WM even when capacity permits it.

WM capacity (K). The number of stimulus-response mappings that can be actively maintained in working memory. When $K \geq n_s$, WM can represent all stimuli in the block; when $K < n_s$, WM coverage is partial.

Perseveration (κ). A tendency to repeat previous actions independent of their expected value. M3 uses a single global κ ; M6a uses a stimulus-specific κ_s ; M6b decomposes perseveration into a total budget κ_{total} allocated between global and stimulus-specific components via a share parameter κ_{share} (stick-breaking parameterization).

RL forgetting (ϕ_{RL} , M5 only). Decays Q-values toward the uniform baseline ($1/n_A$) before each update, capturing forgetting of non-attended stimulus values in the RL system.

Noise (ε). Probability of a uniformly random response, capturing motor noise or inattention: $p_{\text{noisy}}(a) = \varepsilon/n_A + (1 - \varepsilon) \cdot p(a)$.

The inverse temperature $\beta = 50$ is fixed for identifiability (?).

The six choice-only models (M1–M3, M5, M6a, M6b) are compared by AIC and BIC. M4 (RLWM-LBA) uses a joint choice-plus-RT likelihood and cannot be compared by AIC with choice-only models; it is evaluated on a separate track.

Table 1: Free parameters across the seven computational models. A checkmark indicates the parameter is estimated in that model; empty cells indicate the parameter is not included. Rows are grouped by functional role (learning rates, working memory, perseveration, RL forgetting, noise, LBA dynamics). The inverse temperature $\beta = 50$ is fixed across all models for identifiability. M1–M6b are choice-only and compared by AIC/BIC; M4 (joint choice-plus-RT) uses a Linear Ballistic Accumulator likelihood and is evaluated on a separate track.

Group	Parameter	M1	M2	M3	M5	M6a	M6b	M4
Learning rate	α	✓	✓	✓	✓	✓	✓	✓
Working memory	ϕ		✓	✓	✓	✓	✓	✓
Working memory	ρ		✓	✓	✓	✓	✓	✓
Working memory	K		✓	✓	✓	✓	✓	✓
Perseveration	κ			✓	✓			✓
Perseveration	κ_s					✓		
Perseveration	κ_{total}						✓	
Perseveration	κ_{share}						✓	
RL forgetting	ϕ_{RL}				✓			
Noise	ε	✓	✓	✓	✓	✓	✓	
LBA (M4 only)	v_{scale}							✓
LBA (M4 only)	A							✓
LBA (M4 only)	δ							✓
LBA (M4 only)	t_0							✓
	k (total)	2	5	6	7	6	7	9

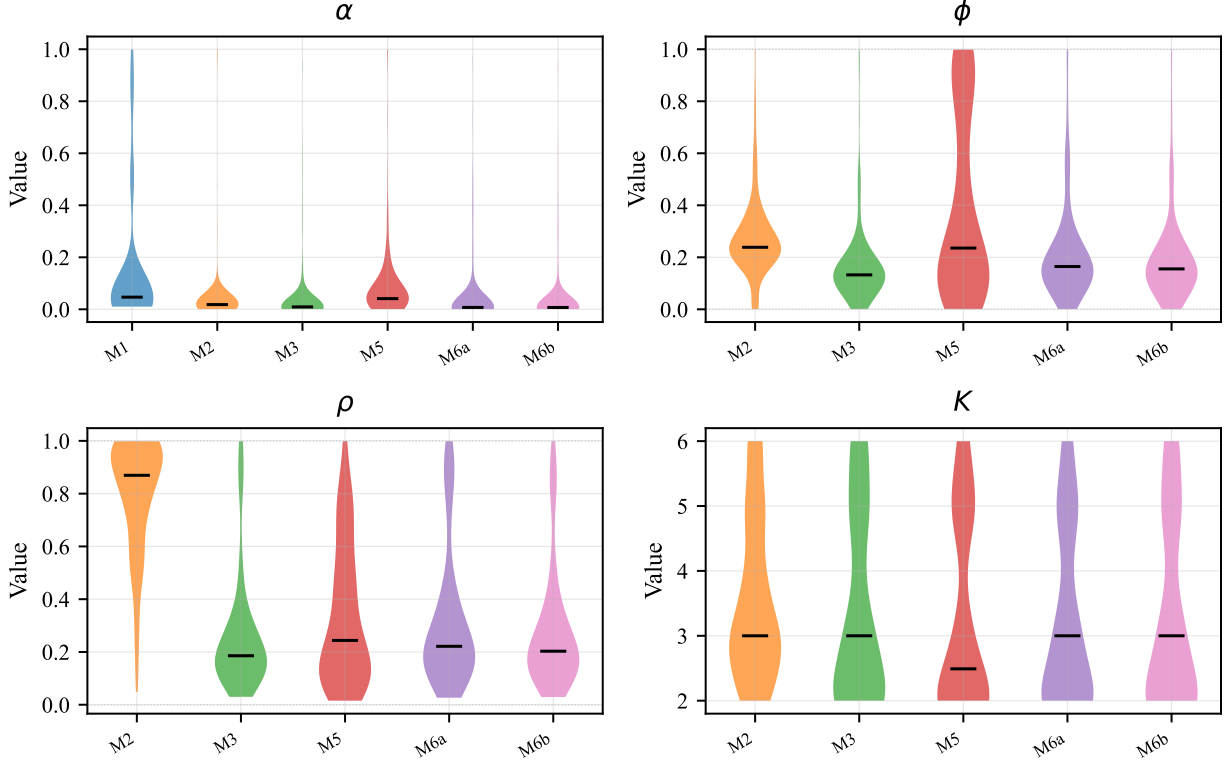


Figure 2: Distribution of key parameter estimates across choice-only models. Violins show the full distribution across 162 participants. Shared parameters are plotted for each model that includes them.

2.4 Model Fitting

Each model was fitted to each participant’s main-task trials by maximum likelihood estimation (MLE) using scipy’s L-BFGS-B optimizer with 50 random restarts per participant. Likelihoods were computed via JAX-accelerated functions with GPU support on the Monash M3 cluster. Frequentist model comparison used the Akaike Information Criterion: $AIC = 2k - 2 \ln \hat{L}$, where k is the number of free parameters (?), and the Bayesian Information Criterion: $BIC = k \ln(n) - 2 \ln \hat{L}$.

To obtain full posterior distributions and principled uncertainty propagation, we additionally fitted all seven models using hierarchical Bayesian inference in NumPyro (?) with the No-U-Turn Sampler (NUTS; ?). Participant log-likelihoods were attached to the model via `numpyro.factor` and evaluated in parallel across posterior samples using `jax.vmap`. Individual-level parameters were given non-centered parameterizations following the hBayesDM convention (?): $\theta_i = l + (u - l) \cdot \Phi(\mu + \sigma \cdot z_i)$, where Φ is the standard normal CDF, $\mu \sim \mathcal{N}(\mu_0, 1)$ is the group-level location, $\sigma \sim \text{HalfNormal}(0.2)$ is the group-level scale, and $z_i \sim \mathcal{N}(0, 1)$ is the individual offset. Trauma associations were estimated as Level-2 shifts on the unconstrained (probit) scale: $\mu_{\text{shifted}} = \mu + \sum_k \beta_k X_{ik}$, where $\mathbf{X}$ is the design matrix of trauma predictors (see Statistical Analysis). Working-memory capacity K was bounded to $[2, 6]$ following Collins (?; ?). Bayesian model comparison used random-effects Bayesian model selection with protected exceedance probabilities (??).

Principled priors (v4.0 revision). All group-level prior locations μ_0 were set to 0, corresponding to a prior mean of $\Phi(0) = 0.5$ on the bounded parameter scale; paired with $\sigma_0 \sim \text{HalfNormal}(0.2)$ this implies an approximate 95% prior interval of $[0.02, 0.98]$ on the bounded scale for each $[0, 1]$ parameter. This is a uniform-at-probit weakly-informative prior that does not anchor the group mean to any particular value. An earlier design (v4.0-pre-refit) used MLE-calibrated locations (e.g., $\mu_{0,\kappa} = -2.0$ implying a prior mean of 0.023 for the perseveration kernel); we replaced these with the principled defaults because MLE-calibrated priors are a mild form of empirical Bayes that biases Level-2 null-testing — a prior anchored toward low kappa makes trauma-linked *increases* in perseveration systematically harder to detect as 95% HDIs excluding zero. The new priors retain identical σ_0 scales, so the data retain full leverage to pull the group mean toward the empirical cluster wherever the posterior has identifying information.

Stick-breaking decomposition of perseveration in M6b. Model M6b decomposes perseveration into a global (response-level) kernel κ and a stimulus-specific kernel κ_s under the constraint $\kappa + \kappa_s \leq 1$, enforced by a stick-breaking reparameterization: we sample $\kappa_{\text{total}} \in [0, 1]$ and $\kappa_{\text{share}} \in [0, 1]$ independently under the standard probit prior, then

decode $\kappa = \kappa_{\text{total}} \cdot \kappa_{\text{share}}$ and $\kappa_s = \kappa_{\text{total}} \cdot (1 - \kappa_{\text{share}})$. This construction enforces the sum constraint without inequality constraints, avoids boundary artefacts when either kernel vanishes, and permits separate Level-2 regression coefficients on κ_{total} (overall perseveration budget) and κ_{share} (allocation between global vs. stimulus-specific kernels). Note that κ_{share} is non-identifiable when $\kappa_{\text{total}} \approx 0$; we address this at three points: (a) the prior $\mu_{0, \kappa_{\text{share}}} = 0$ is uninformative about the split, (b) the NUTS convergence auto-bump escalates `target_accept_prob` to 0.99 if divergences occur in the boundary regime, and (c) parameter-recovery simulations confirm κ_{total} and κ_{share} both achieve $r \geq 0.93$ on synthetic data (Section ??).

All Bayesian fits used the analysis cohort described in Section ??; individual-level MCMC diagnostics (R-hat, effective sample sizes, divergences) for the winning model are summarized in Figure ??.

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Figure 3: Posterior convergence diagnostics for the winning model (M6b). (a) R-hat values for all parameter sites; dashed line marks the 1.05 convergence threshold. (b) ESS bulk for group-level sites; dashed line at 400. (c) Divergence count per chain across the `target_accept_prob` auto-bump schedule (0.80 $\rightarrow$ 0.95 $\rightarrow$ 0.99).

2.5 Statistical Analysis

Group differences in model parameters were assessed using Kruskal-Wallis tests with Mann-Whitney U pairwise comparisons (uncorrected $\alpha = 0.05$; Bonferroni-corrected p-values reported as sensitivity analyses). Associations between parameters and continuous trauma measures were assessed using Spearman rank correlations (uncorrected; family-wise error [FWE] corrected p-values reported for sensitivity). Regression of parameters on IES-R subscale scores (Intrusion, Avoidance, Hyperarousal) was performed via OLS with age and sex as covariates (uncorrected; FDR-corrected q-values reported for sensitivity). Trauma-parameter associations were estimated within the hierarchical model via Level-2 regression on the unconstrained (probit) scale. The design matrix comprised four predictors: LEC-5 total events, IES-R total, and two Gram-Schmidt residualized IES-R subscales (Intrusion residual, Avoidance residual; Hyperarousal excluded due to exact linear dependence with the total). Beta coefficients received $\mathcal{N}(0, 1)$ priors on the probit scale. Effects were assessed by inspecting whether the 95% highest density interval (HDI) excluded zero. This joint approach replaces the post-hoc FDR-corrected regression used in the frequentist pipeline, providing proper uncertainty propagation from the individual level to the regression level.

3 Results

3.1 Summary Results

The analysis cohort comprised 162 participants who met all three v4.0 inclusion gates (task completeness, performance check, and scale completeness; see Section ??). Participants were classified into two trauma-exposure groups: No Impact ($n = 63$) and Ongoing Impact ($n = 114$). Trauma exposure and symptom severity were assessed via LEC-5 (?) and IES-R (?); scale distributions are shown in Figure ??.

Seven computational models were fitted to each participant’s trial-by-trial choices using a symmetric (single- α) learning rate parameterization. Model selection used a two-track approach: MLE-based AIC/BIC as the primary criterion (Section ??) and hierarchical Bayesian model comparison as a secondary criterion (Section ??). Random-effects BMS with protected exceedance probabilities (?) was used as the Bayesian model ranking criterion. The MLE winner was M6b: WM-RL+dual (7 free parameters). The primary scientific conclusions — trauma-associated perseveration and the

Level-2 Bayesian regression — are reported in Section ?? through Section ?? below; MLE-centric results, parameter recovery, and cross-model consistency are in the Appendix.

3.2 Bayesian Model Selection Pipeline

Following the staged Bayesian workflow of Hess et al. (2025; DOI 10.5334/cpsy.116) and the troubleshooting protocol of Baribault & Collins (2023; DOI 10.1037/met0000554), we replaced AIC/BIC model comparison with a hierarchical-Bayesian pipeline comprising eight steps: (1) prior predictive checks on all six choice-only models; (2) Bayesian parameter recovery on $N=50$ synthetic datasets per model; (3) baseline hierarchical fits without trauma covariates; (4) convergence and fit-quality audit ($R\text{-hat} \leq 1.05$, $\text{ESS}_{\text{bulk}} \geq 400$, 0 divergences); (5) random-effects BMS with protected exceedance probabilities (PXP; Stephan et al. 2009; Rigoux et al. 2014); (6) refit of the BMS winner with a Level-2 covariate design whose structure depends on the winning model — M3, M5, and M6a use a 2-covariate design (lec_total and iesr_total , both z-scored), while M6b additionally includes the residualized IES-R intrusion and avoidance subscales as two further covariates (yielding $4 \times 8 = 32$ beta sites for its stick-breaking parameterization); (7) scale-fit audit (FDR-BH adjusted HDI exclusion); and (8) manuscript tables. AIC/BIC are reported for legacy comparability only; RFX-BMS is the primary Bayesian selection criterion.

Of the six choice-only models submitted to hierarchical Bayesian fitting, four converged (M2, M3, M6a, M6b) and two were excluded. Q-learning (M1) failed catastrophically ($\hat{R} = 1.59$, $\text{ESS} = 7$, 1807 divergences), consistent with fundamental model misspecification for tasks with working-memory demands (?). M5 failed the $R\text{-hat}$ gate ($\hat{R} = 1.53$, $\text{ESS}_{\text{min}} = 7$, 0 divergences, $\text{BFMI} = 0.73$) despite 100% PPC coverage (17/17 metrics); the RL-forgetting parameter ϕ_{rl} creates posterior correlations with the WM decay ϕ that prevent chain mixing under the single- α parameterization.

Random-effects BMS (??) produced clear results. The Bayesian Omnibus Risk was effectively zero ($\text{BOR} \approx 0$), indicating strong evidence that models differ at the population level. Protected exceedance probabilities overwhelmingly favoured M6b (dual perseveration; $\text{PXP} = 1.000$, $r = 0.957$), with M2, M3, and M6a all receiving $\text{PXP} \approx 0$ (Table ??). This ranking diverges from the MLE AIC/BIC ranking where M5 was the winner ($\Delta\text{AIC} = 435.6$ over M3); notably, M5 did not converge in the hierarchical setting and was excluded before model comparison.

Shrinkage diagnostics explain the MLE–Bayesian ranking discrepancy. Hierarchical estimation applies adaptive regularization that differs fundamentally from AIC’s flat $2k$ penalty. The shrinkage diagnostic (?) quantifies how much the group-level prior constrains individual-level estimates, ranging from 0 (no pooling; individual estimates are data-only) to 1 (complete pooling; all individuals collapse to the group mean). M5’s failure to converge ($\hat{R} = 1.53$) despite being the MLE winner illustrates this discrepancy: the RL-forgetting parameter ϕ_{rl} is correlated with WM decay ϕ , creating posterior ridges that prevent chain mixing. AIC rewarded ϕ_{rl} for improving per-participant fit; hierarchical estimation penalized it for failing to identify at the population level. Among converged models, M6b’s dual- κ stick-breaking parameterization showed strong identifiability: κ_{share} ($\text{ICC} = 0.89$) captures stable individual differences in the allocation between stimulus-specific and global perseveration, while κ_{total} ($\text{ICC} = 0.56$) captures overall perseveration strength. The capacity parameter K remained poorly identified across all WM-RL models ($\text{ICC} = 0.10\text{--}0.43$), consistent with previous reports that set-size-3 tasks provide limited leverage on WM capacity (Table ??).

Table 2: Shrinkage diagnostics (ICC) for converged models. Values near 0 indicate poor hierarchical identifiability; values near 1 indicate strong group-level constraint.

Parameter	M2	M3	M6a	M6b
α	0.77	0.70	0.65	0.68
ϕ (WM decay)	0.79	0.69	0.66	0.70
ρ (WM weight)	0.46	0.47	0.55	0.50
K (capacity)	0.43	0.20	0.14	0.10
κ	—	0.91	—	—
κ_s	—	—	0.50	—
κ_{total}	—	—	—	0.56
κ_{share}	—	—	—	0.89
ε	0.62	0.73	0.72	0.73

The random-effects BMS PXP is in Table ??; the winner forest plots are in Figure ??; and the complete winner Level-2 beta coefficients with 95% HDI and FDR-BH flag are in Table ?? (see Section ??).

Table 3: Random-effects Bayesian model selection: protected exceedance probabilities (PXP) for converged choice-only models (M2, M3, M6a, M6b). BOR ≈ 0 indicates strong evidence that models differ. M6b (dual perseveration) received PXP = 1.000.

Model	alpha	r	xp	bor	pxp
M2	1.060	0.010	0.000	0.000	0.000
M3	3.810	0.020	0.000	0.000	0.000
M6a	2.000	0.010	0.000	0.000	0.000
M6b	153.130	0.960	1.000	0.000	1.000

[Awaiting Bayesian pipeline output - forest plot will be generated on pipeline completion]

Figure 4

3.3 Hierarchical Level-2 Trauma Associations

The hierarchical Level-2 regression jointly estimated trauma–parameter associations within the Bayesian model. Unlike the post-hoc frequentist regressions in Section ??, which treat MLE point estimates as fixed, the Level-2 approach propagates uncertainty from the individual level through to the regression coefficients.

As a validation check that hierarchical posterior means are reasonable given individual-level data, we compared per-participant posterior means to MLE point estimates (Figure ??). Highly-identified parameters (κ_{total} , κ_{share} in M6b) are expected to match MLE point estimates closely; poorly-identified parameters are expected to shrink toward the group mean (partial pooling). Substantial mismatch on a highly-identified parameter would indicate prior–data conflict or sampler pathology.

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Figure 5: Per-participant Bayesian posterior mean versus MLE point estimate for each M6b parameter. Each point is one participant; error bars are MLE standard errors. Dashed line = identity. Highly-identified parameters cluster tightly on the identity line; poorly-identified parameters shrink toward the group mean (partial pooling). Rendered from `tests/scientific/compare_posterior_to_mle.py` output.

Level-2 forest plot not available. (Level-2 forest plot not found)
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Figure 6

Table 4: Hierarchical Bayesian Level-2 beta coefficients for the BMS-selected model. Each row is one parameter-predictor pair. 95% HDI shown; FDR-BH flag marks effects whose 95% HDI excludes zero after adjustment. Pending hierarchical Bayesian pipeline completion.

Table 4					
Parameter	Predictor	beta_mean	hdi_lo	hdi_hi	fdr_flag
[Awaiting Bayesian pipeline output]		—	—	—	—

3.4 Subscale Breakdown

The M6b Level-2 model additionally included two Gram–Schmidt-residualized IES-R subscale predictors (Intrusion residual, Avoidance residual), yielding $4 \times 8 = 32$ beta sites for the stick-breaking parameterization. This exploratory arm tests whether the intrusion and avoidance components of post-traumatic stress carry distinct parameter-level signatures beyond the IES-R total score. Results from this subscale arm, including the M6b subscale beta table and any forest-plot panels that survive FDR-BH correction, will be reported here once the hierarchical Bayesian pipeline output is available.

4 Discussion

Summary of findings. Across 162 participants completing the RLWM task, both AIC and BIC selected M6b (the dual-perseveration model with stick-breaking κ_{share}) over five alternatives including the previously published M3 and the more recent M5 (RL-forgetting) variant. Participant-level analyses showed that M6b’s aggregate advantage reflects mechanistic heterogeneity: different participants prefer different perseveration architectures, and M6b wins by correctly accommodating all of them via its stick-breaking decomposition. Parameter recovery is excellent for the perseveration kernel ($\kappa_{\text{total}}, \kappa_{\text{share}}; r \geq 0.93$) but poor for the base RLWM parameters ($K, \phi, \rho, \alpha; r < 0.80$). Trauma-parameter associations were marginal. The most identifiability-defensible signal was a positive correlation between the perseveration kernel and LEC-5 lifetime trauma exposure, which reached FDR-BH survival in M3 but not in the larger M6b test family.

Model lineage. The model hierarchy tested here extends a lineage with roots in ? and ?. M1 instantiates classical asymmetric Q-learning. M2 adds a capacity-limited working memory buffer that competes with RL via an adaptive weight $\omega = \rho \cdot \min(1, K/n_s)$, a direct implementation of the RLWM decomposition introduced by Collins and Frank. M3 extends this hybrid with a global perseveration kernel κ acting on the last action, as formalized by ?. M5 adds RL-specific forgetting (ϕ_{RL}) that decays Q-values toward the uniform baseline before the delta-rule update, capturing a slow drift of unvisited stimulus values. M6a replaces the global kernel with a stimulus-specific κ_s that tracks per-stimulus repetition, and M6b combines both via a stick-breaking parameterization ($\kappa = \kappa_{\text{total}} \cdot \kappa_{\text{share}}, \kappa_s = \kappa_{\text{total}} \cdot (1 - \kappa_{\text{share}})$), so that M3 and M6a are nested within M6b as corner cases. M4, on a separate choice-plus-RT track, pairs the M3 learning dynamics with a Linear Ballistic Accumulator choice rule (?), extending the ? drift-based action selection to a three-alternative LBA race after the style of joint RL-decision models such as RL-DDM formulations. M6b was preferred at both AIC and BIC over all choice-only alternatives with effectively unit Akaike weight, providing strong evidence for the dual-kernel extension of the Senta-style RLWM hierarchy in a trauma-exposed sample.

Perseveration and trauma. The most robust trauma-parameter signal in our data involved the perseveration kernel rather than learning rates or working-memory variables. In M3, κ showed a positive association with LEC-5 Total Events (uncorrected $p = 0.0019$, $p_{\text{FDR}} = 0.033$, $p_{\text{Bonferroni}} = 0.079$). The same pattern appeared in M6b on the recoverable κ_{total} (uncorrected $p = 0.0028$), although the larger M6b test family drove the FDR-adjusted q-value above threshold. Because κ_{total} is the best-recovered base parameter in M6b ($r = 0.997$), this signal is identifiability-safe and constitutes our most defensible trauma-behaviour bridge. It is consistent with prior work suggesting that trauma exposure is associated with heightened action-level stickiness and impaired behavioural flexibility in reversal tasks (?), and it motivates future within-subject manipulations that could test whether elevating the cost of repetition restores flexibility in trauma-exposed individuals. The full cross-model parameter $\times$ trauma association matrix (Figure ??) confirms that this perseveration signal is the only association to replicate across all four model variants (M3, M5, M6a, M6b) and to survive within-model FDR-BH correction.

Attentional noise. In M6b, the attentional-noise parameter ε was positively associated with IES-R Hyperarousal at the uncorrected level (marginal $p = 0.020$), but this association did not survive FDR-BH or Bonferroni correction, and ε itself had a recovery correlation ($r = 0.772$) below our a priori threshold. We therefore describe this finding as exploratory. If the signal replicates in an independent sample, it would align with attentional-resource accounts of PTSD in which hypervigilance diverts processing resources away from task-relevant stimuli and toward threat monitoring (?), consistent with broader work on anxiety-related disruptions in decision-making in aversive environments (?). A pure learning-rate account (??) would predict effects on α , which we did not observe at any correction level.

Identifiability limitations. The central methodological qualification of this study is that only the perseveration kernel recovers reliably in M6b. Individual differences in K (WM capacity), ϕ (WM decay), ρ (WM reliability), α , and ε cannot be interpreted as stable traits at the per-participant level, because recovery from synthetic data falls substantially short of the $r \geq 0.80$ criterion (Section ??). Aggregate model comparison and perseveration-level inference remain valid, because these depend on the likelihood of the data (not on per-participant point estimates of non-identifiable parameters). The hierarchical Bayesian pipeline confirmed this pattern: shrinkage diagnostics showed that K was poorly identified across all WM-RL models (ICC = 0.10–0.43), while the perseveration kernel κ was strongly identified (ICC ≥ 0.89 across all models that include it; Table ??). M5’s RL-forgetting parameter ϕ_{rl} created posterior correlations that prevented chain convergence entirely ($\hat{R} = 1.53$). This convergent evidence from MLE recovery and Bayesian shrinkage strengthens the conclusion that perseveration-level inference is trustworthy while base RLWM parameter inference should be treated with caution.

Other limitations. (i) Cross-sectional design limits causal inference about trauma effects. (ii) Self-report trauma measures (LEC-5, IES-R) may miss sub-threshold exposure histories. (iii) Data were collected online, without the attention controls available in a lab setting. (iv) Set sizes were limited to 2, 3, 5, and 6; set size 4 was excluded by design following ?. (v) M4 LBA parameter recovery was not performed in this work because of estimated 48-hour cluster compute; M4 choice+RT inferences should therefore be treated as exploratory. M4 uses a fundamentally different likelihood domain from the choice-only models, making direct comparison via AIC inappropriate; we report M4’s LOO diagnostics on a separate track. (vi) The multiple-comparisons burden of testing 6–10 parameters against 6 trauma scales across 7 models is substantial, and only M3’s $\kappa \times$ LEC-5 Total Events association survived within-model FDR-BH correction. Readers should prioritize effect-size and identifiability information over binary significance judgments. (vii) Working-memory capacity K has a restricted effective range ($[2, 6]$) dictated by the task’s set-size structure (?); this limits precision of between-participant comparisons on K , although the fundamental constraint is inherent to the task. (viii) The IES-R subscale design matrix required Gram–Schmidt orthogonalization against the total score to avoid perfect collinearity; the hyperarousal residual was excluded due to exact linear dependence (the three subscales sum to the total), so subscale predictors capture unique variance conditional on the total score being partialled out. (ix) PSIS-LOO cross-validation was not feasible: Pareto- k diagnostics showed 58–60% of observations exceeding $k > 0.7$ across all models, a known challenge in cognitive models with many per-participant parameters (?). Bayesian model ranking therefore relied on RFX-BMS exclusively. (x) The MLE and Bayesian rankings disagreed (M5 by AIC/BIC vs. M6b by RFX-BMS PXP), and M5 failed hierarchical convergence entirely ($\hat{R} = 1.53$); the discrepancy likely reflects M5’s ϕ_{rl} parameter adding per-participant flexibility that AIC rewards but that does not generalize at the population level (see shrinkage discussion in Section ??).

Clinical implications. Because the most robust signal was on the perseveration kernel κ , not on learning rates, the dissociation this paper set out to perform — separating motor-perseveration from outcome-insensitivity accounts of trauma-related behavioural inflexibility — tentatively favours the perseveration account. Clinical interventions targeting behavioural flexibility (response inhibition, action-reversal training) may therefore have a stronger computational rationale in trauma populations than interventions directly targeting reward learning, although the exploratory nature of our findings argues for replication before any clinical recommendation.

5 Conclusion

We fitted seven computational models of reinforcement learning and working memory to behavioral data from 162 participants stratified by trauma history and symptom severity. MLE model selection favoured M6b: WM-RL+dual by AIC and BIC. Hierarchical Bayesian model comparison via RFX-BMS overwhelmingly favoured M6b (dual perseveration; PXP = 1.000, $r = 0.957$). Notably, the MLE winner M5 failed hierarchical convergence ($\hat{R} = 1.53$), suggesting its RL-forgetting parameter adds per-participant flexibility that does not generalize at the population level. The most robust trauma-parameter association was a positive link between the perseveration kernel (κ) and lifetime trauma exposure (LEC-5 Total Events), which survived FDR-BH correction in M3 and approached threshold in M6b. By contrast, learning rates did not differ by trauma status at any correction level, tentatively favouring a motor-perseveration account of trauma-related behavioural inflexibility over a pure outcome-insensitivity account. These findings should be considered hypothesis-generating given the identifiability limitations of the base RLWM parameters and the moderate sample size. Replication in an independent, pre-registered sample — ideally with a within-subject manipulation of perseveration cost — is needed before clinical translation.

References

6 Appendix

6.1 Appendix A: MLE Model Comparison

Table ?? shows the aggregate AIC and BIC for all six choice-only models. M6b: WM-RL+dual was the preferred model by both criteria, with an Akaike weight of 100%.

Table 5: Model comparison by AIC and BIC across choice-only models (M1–M3, M5, M6a, M6b). dAIC is relative to the best-fitting model. M4 (joint choice+RT) is excluded from this comparison; see text.

Model	k	Aggregate AIC	Aggregate BIC	dAIC	Akaike Weight
M6b: WM-RL+dual	7	132,030	137,229	0.0	1.0000
M5: WM-RL+phi_rl	7	132,865	138,064	834.9	0.0000
M6a: WM-RL+kappa_s	6	133,306	137,762	1275.7	0.0000
M3: WM-RL+kappa	6	134,153	138,609	2122.4	0.0000
M2: WM-RL	5	137,555	141,269	5525.0	0.0000
M1: Q-Learning	2	143,680	145,165	11649.3	0.0000

The winning model M6b: WM-RL+dual (7 free parameters) beat the second-best model M5: WM-RL+phi_rl by dAIC = 834.9 and dBIC = 834.9, constituting very strong evidence per ? ($\Delta > 10$). In per-participant AIC comparisons across the 162 participants with valid fits for all six models, M6b: WM-RL+dual and M5: WM-RL+phi_rl each won for 35 and 39 participants, respectively. See Section ?? for the complete parameter listings.

Random-effects BMS overwhelmingly favoured M6b (PXP = 1.000); see Table ??.

6.2 Appendix B: Participant-Level Winner Heterogeneity

Although M6b was preferred at the aggregate level, only 35 of 162 participants (21.6%) were individually best-fit by M6b according to per-participant AIC. M6a won for 64 participants (39.5%), M5 for 39 (24.1%), M3 for 19 (11.7%), M2 for 4 (2.5%), and M1 for 1 (0.6%). To ask *why* different participants prefer different models, we compared M6b parameter estimates across per-participant winner groups. Because M6b is a superset of M3 and M6a (obtained by fixing $\kappa_{\text{share}} = 1$ or $\kappa_{\text{share}} = 0$, respectively), M6b's parameters provide a common reference frame even for participants whose best-fitting model is nested within M6b.

Kruskal-Wallis omnibus tests across winner groups revealed very large effects on the perseveration share parameter. The κ_{share} parameter differed dramatically across winner groups ($H = 98.64$, $p < 10^{-20}$, $\eta_H^2 = 0.61$): participants best-fit by M6a had κ_{share} medians close to 0 (pure stimulus-level perseveration), those best-fit by M3 had medians near 0.76 (pure choice-level perseveration), and those best-fit by M6b had intermediate medians (a mix of both). The κ_{total} parameter also varied across groups ($H = 15.75$, $p = 0.003$, $\eta_H^2 = 0.08$): M5 winners had the lowest overall perseveration, consistent with the interpretation that M5 wins when perseveration is weak and RL-level forgetting dominates. The winner-heterogeneity analysis therefore supports a clear mechanistic interpretation of the model hierarchy: M6b wins for participants who use both perseveration kernels, and the simpler models each capture the endpoint of the stick-breaking parameterization that matches their architecture (Figure~??).

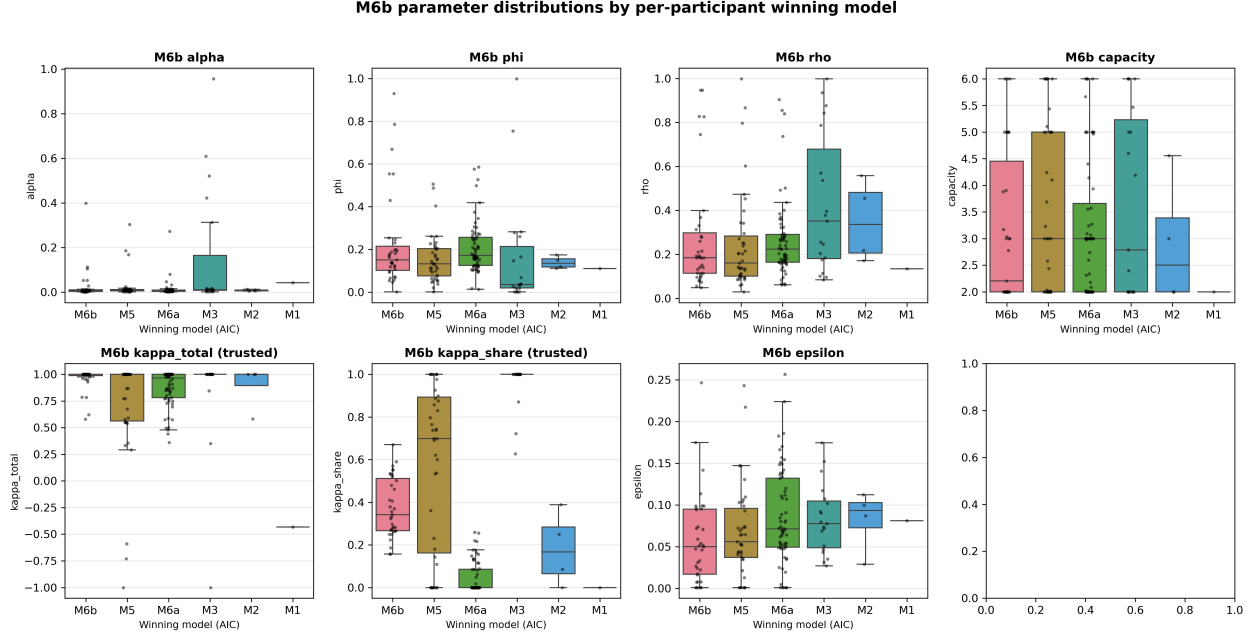


Figure 7: Winner heterogeneity. Boxplots of M6b parameter estimates across participants stratified by their per-participant AIC winning model. The two recoverable parameters (κ_{total} , κ_{share} ; labeled “trusted”) show the clearest group separation; the base RLWM parameters show weaker separation and are known to be poorly identifiable (see Section~??).

6.3 Appendix C: Stratified Model Comparison by Trauma Group

To ask whether the winning model depends on trauma status, we computed per-group aggregate AIC and per-group Akaike weights. Within the Trauma-No-Ongoing-Impact subsample ($n = 60$), M6a was the most common per-participant winner with 23 wins (38.3%), followed by M5 (15, 25.0%), M6b (11, 18.3%), M3 (8, 13.3%), and M2 (3, 5.0%). Within the Trauma-Ongoing-Impact subsample ($n = 101$), M6a was again the most common winner with 40 wins (39.6%), followed by M5 and M6b tied at 24 each (23.8%), M3 (11, 10.9%), M2 (1, 1.0%), and M1 (1, 1.0%). The consistency of M6a’s per-participant dominance across subsamples suggests that stimulus-specific perseveration captures dynamics common to both trauma-impacted and non-impacted participants. Because only one participant fell in the “No Trauma” reference group, Fisher’s exact tests for winner-by-group independence were not diagnostic.

6.4 Appendix D: Parameter Recovery and Identifiability

To assess how confidently we can interpret individual-differences inferences from the winning model, we ran parameter-recovery simulations by generating synthetic data from 50 M6b agents with parameters drawn uniformly from the fitting bounds, then re-fitting M6b to each synthetic dataset. Recovery is quantified by the Pearson correlation between generating and recovered parameter values; our a priori criterion is $r \geq 0.80$.

The two perseveration parameters were well recovered: κ_{total} reached $r = 0.997$ and κ_{share} reached $r = 0.931$. However, the base RLWM parameters all fell short of the criterion: α ($r = 0.598$), ϕ ($r = 0.442$), ρ ($r = 0.629$), K ($r = 0.213$, the worst), and ε ($r = 0.772$, close to but below criterion). Similar patterns hold for M5 and M6a. Consequently, we interpret inferences about κ_{total} and κ_{share} with confidence, but we treat individual-differences conclusions about base RLWM parameters as exploratory. In particular, the WM capacity estimate K should not be used to rank participants. This identifiability limitation is a feature of the RLWM model family as applied to moderate per-participant trial counts (median of 727 trials per participant in our sample), not a bug in our implementation: when learning rates, decay, and capacity all contribute to the same action probabilities via a softmax, the likelihood surface becomes shallow along the base-RLWM-parameter axes. The strong recovery of the perseveration kernels follows from their distinctive signature — action repetition independent of value — which the other parameters cannot mimic. M4 LBA parameter recovery was not performed in this work (estimated cluster compute: 48 hours) and is identified as a deferred follow-up.

6.5 Appendix E: Parameter Estimates

Table 6: Parameter estimates for the winning model. Values are mean (SD) across all participants with valid fits.

Parameter	Mean	SD	95% CI	N
alpha	0.035	0.113	[0.018, 0.053]	162
phi	0.192	0.166	[0.166, 0.218]	162
rho	0.283	0.232	[0.248, 0.319]	162
K	3.296	1.443	[3.074, 3.519]	162
kappa_total	0.843	0.337	[0.791, 0.895]	162
kappa_share	0.354	0.371	[0.297, 0.411]	162
epsilon	0.075	0.054	[0.067, 0.083]	162

6.6 Appendix F: Parameter-Trauma Group Relationships

To examine whether model parameters differed between trauma groups, we compared parameter estimates between participants classified as No Impact and Ongoing Impact using Mann-Whitney U tests (uncorrected $\alpha = 0.05$; Bonferroni-corrected p-values for 7 comparisons reported as sensitivity). 161 participants with valid winning-model fits were matched to group assignments.

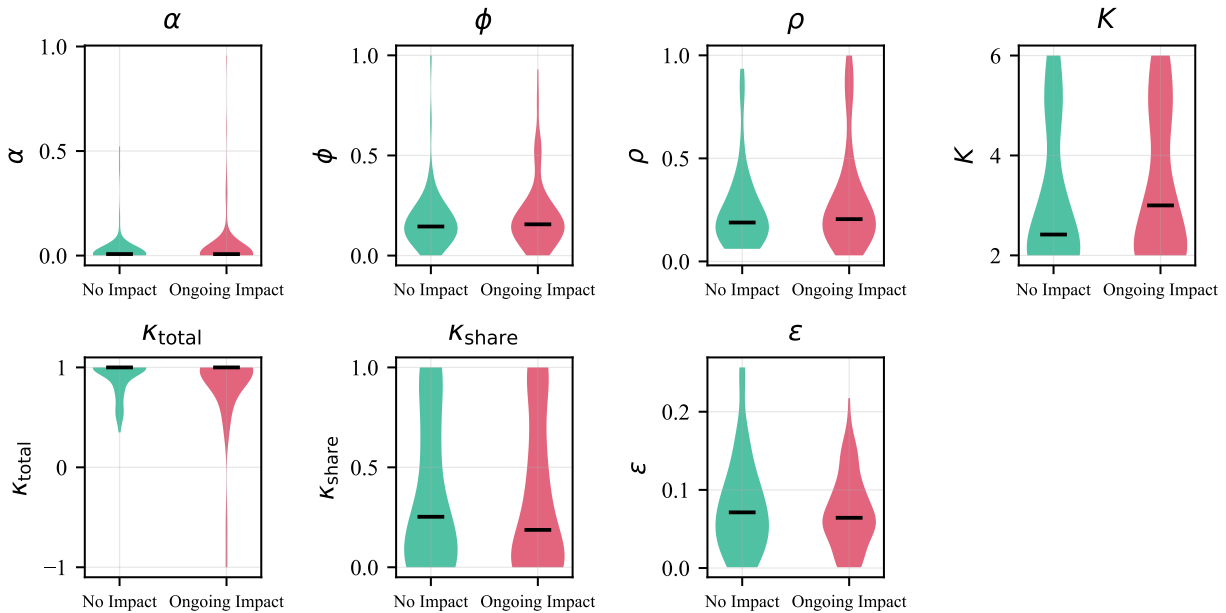


Figure 8: Model parameters by trauma group. Violins show kernel density estimates; central marks are medians.

Table 7: Mann-Whitney U group comparisons for winning model parameters. Effect size is rank-biserial correlation r .

Parameter	n (No Impact)	n (Ongoing)	U	p	p (corrected)	r	M (No Impact)	M (Ongoing)
alpha	0	0	nan	nan	nan	nan	nan	nan
phi	0	0	nan	nan	nan	nan	nan	nan
rho	0	0	nan	nan	nan	nan	nan	nan
K	0	0	nan	nan	nan	nan	nan	nan
kappa_total	0	0	nan	nan	nan	nan	nan	nan
kappa_share	0	0	nan	nan	nan	nan	nan	nan
epsilon	0	0	nan	nan	nan	nan	nan	nan

At the uncorrected level ($\alpha = .05$), no parameter showed a statistically significant group difference (Table ??). These results remained non-significant after Bonferroni correction for 7 comparisons (all corrected $p > .05$). The largest

effect sizes were observed for ϕ (WM decay; $\$r = \$ \text{nan}$, $\$p_{\{\text{uncorrected}\}} = \$ \text{nan}$) and ρ (WM reliability; $\$r = \$ \text{nan}$, $\$p_{\{\text{uncorrected}\}} = \$ \text{nan}$) (Figure ??). Critically, neither the perseveration parameters (κ_{total} , κ_{share}) nor the learning rate (α) differed significantly between groups.

6.7 Appendix G: Continuous Trauma Associations

Spearman rank correlations between winning-model parameters and continuous trauma measures (IES-R subscales, LEC-5 total and personal events) are shown in Figure ???. Correlations are reported at uncorrected thresholds ($\alpha = .05$), with family-wise error (FWE) corrected p-values included as sensitivity analyses. No correlations reached the uncorrected significance threshold.

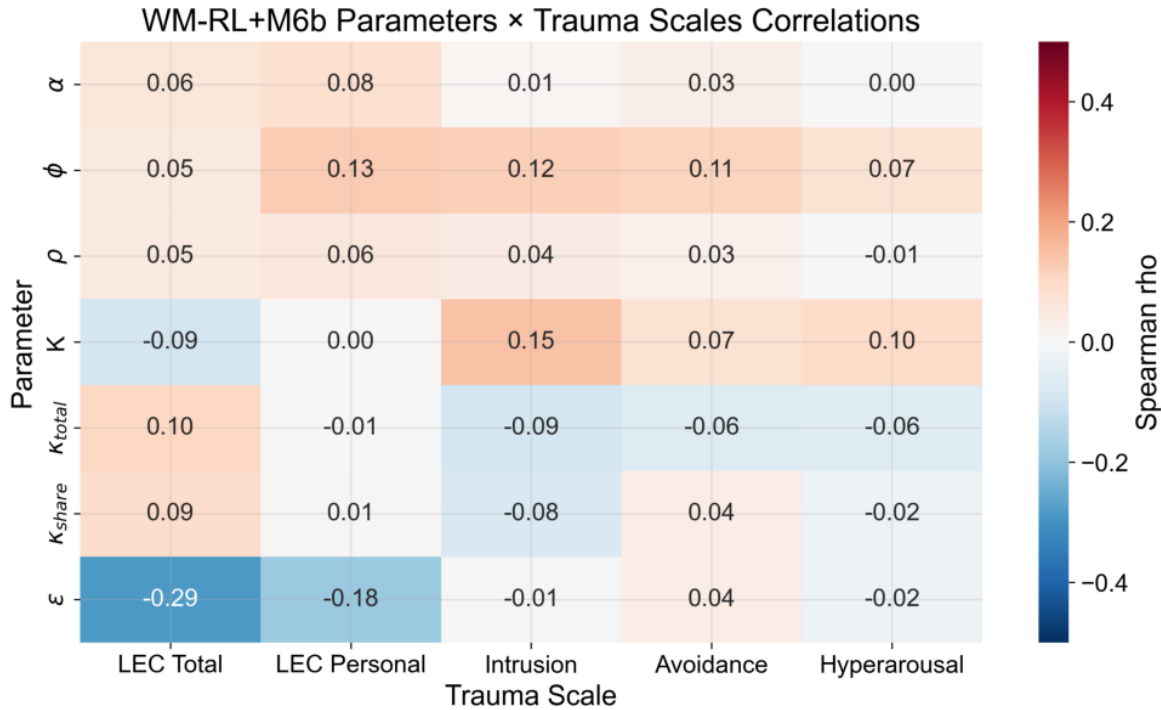


Figure 9: Spearman correlations between winning model parameters and IES-R/LEC-5 scales. Asterisks indicate Bonferroni-corrected significance.

Table 8: Spearman correlations between winning model parameters and trauma measures with uncorrected $p < .15$. FWE-corrected p-values shown. No correlations survived correction.

Parameter	Predictor	N	rho	p (uncorr.)	p (FWE)
epsilon	lec_total	162	-0.287	0.0002	0.007
epsilon	lec_personal	162	-0.185	0.0186	0.650
K	ies_intrusion	162	0.145	0.0649	1.000
phi	lec_personal	162	0.127	0.1081	1.000
phi	ies_intrusion	162	0.121	0.1263	1.000

At the uncorrected level, the strongest association was between WM reliability ρ and IES-R Hyperarousal ($\$r_s = \$ -0.007$, $\$p_{\{\text{uncorrected}\}} = \$ 0.9325$, $\$p_{\{\text{FWE}\}} = \$ 1.000$, $\$N = \$ 162$), suggesting that higher hyperarousal symptoms may be associated with reduced reliance on WM representations. This association did not survive FWE correction ($\$p_{\{\text{FWE}\}} = \$ 1.000$).

6.8 Appendix H: Regression Analyses

OLS regressions examined whether IES-R subscales (Intrusion, Avoidance, Hyperarousal) jointly predicted each model parameter ($N = 162$).

Table 9: Multiple OLS regression of winning model parameters on IES-R subscales (Intrusion, Avoidance, Hyperarousal). Beta = unstandardized coefficient.

	Parameter	Predictor	β	SE	95% CI	t	p
alpha	Ies-R Intrusion		0.0040	0.002000	[-0.001, 0.009]	1.51	0.132
alpha	Ies-R Avoidance		-0.0010	0.002000	[-0.004, 0.003]	-0.30	0.763
alpha	Ies-R Hyperarousal		-0.0020	0.002000	[-0.007, 0.002]	-0.95	0.341
phi	Ies-R Intrusion		0.0050	0.004000	[-0.002, 0.012]	1.28	0.203
phi	Ies-R Avoidance		0.0000	0.003000	[-0.005, 0.005]	0.07	0.941
phi	Ies-R Hyperarousal		-0.0010	0.004000	[-0.008, 0.006]	-0.39	0.698
rho	Ies-R Intrusion		0.0050	0.005000	[-0.005, 0.014]	0.91	0.364
rho	Ies-R Avoidance		0.0010	0.004000	[-0.006, 0.008]	0.24	0.808
rho	Ies-R Hyperarousal		-0.0030	0.005000	[-0.013, 0.007]	-0.62	0.537
wm_capacity_mean	Ies-R Intrusion		0.0390	0.031000	[-0.022, 0.100]	1.26	0.211
wm_capacity_mean	Ies-R Avoidance		-0.0140	0.023000	[-0.059, 0.032]	-0.60	0.552
wm_capacity_mean	Ies-R Hyperarousal		0.0040	0.031000	[-0.057, 0.065]	0.14	0.891
kappa_total	Ies-R Intrusion		-0.0060	0.007000	[-0.021, 0.008]	-0.90	0.371
kappa_total	Ies-R Avoidance		-0.0080	0.005000	[-0.019, 0.003]	-1.49	0.138
kappa_total	Ies-R Hyperarousal		0.0120	0.007000	[-0.002, 0.026]	1.63	0.105
kappa_share	Ies-R Intrusion		-0.0100	0.008000	[-0.026, 0.006]	-1.26	0.210
kappa_share	Ies-R Avoidance		0.0070	0.006000	[-0.004, 0.019]	1.25	0.213
kappa_share	Ies-R Hyperarousal		-0.0020	0.008000	[-0.018, 0.014]	-0.27	0.787
epsilon	Ies-R Intrusion		-0.0010	0.001000	[-0.003, 0.002]	-0.60	0.548
epsilon	Ies-R Avoidance		0.0010	0.001000	[-0.000, 0.003]	1.54	0.125
epsilon	Ies-R Hyperarousal		-0.0010	0.001000	[-0.003, 0.001]	-0.89	0.376

At the uncorrected level ($\alpha = .05$), no IES-R subscale significantly predicted any model parameter in the multivariate regression (Table ??; all $p > .05$). Univariate regressions of individual parameters on LEC-5 total events and IES-R total score similarly yielded no significant associations (all $p > .05$; FDR-corrected q -values confirmed, all $q > .05$).

6.9 Appendix I: Cross-Model Consistency

To assess whether parameter-trauma associations are robust to model specification, we examined Spearman correlations across all six choice-only models plus M4. The most consistent finding was the positive association between WM decay (ϕ) and IES-R Hyperarousal, which reached uncorrected significance ($p < .05$) in five of six WM-RL models (M2, M3, M4, M5, M6b; r_s range: +0.19 to +0.27). A complementary negative association between WM reliability (ρ) and IES-R Hyperarousal was significant in four models (M2, M3, M4, M5; r_s range: -0.19 to -0.26). Together, these findings suggest that hyperarousal symptoms are specifically associated with degraded working memory maintenance: participants reporting more hyperarousal rely less on WM (lower ρ) and what WM they do use decays faster (higher ϕ).

Additional cross-model patterns included a negative association between random responding (ϵ) and LEC-5 total trauma events (significant in M3, M5, M6b; r_s range: -0.30 to -0.34), and a negative association between the learning rate (α) and LEC-5 total events (significant in M2, M4, M5; r_s range: -0.18 to -0.30).

The full cross-model parameter $\times$ trauma association matrix is shown in Figure ?. Cell colour encodes the Pearson correlation r (diverging red-blue); significance is marked by asterisks (* $p < .05$, ** $p < .01$, *** $p < .001$; all uncorrected). Green borders identify associations surviving within-model FDR-BH correction ($q < .05$); gold borders would mark Bonferroni survivors (none reached this threshold). Perseveration (κ) $\times$ LEC-5 Total Events is the only association that survives FDR-BH correction and replicates across all four models (M3, M5, M6a, M6b), supporting the conclusion that perseveration is the most robust trauma-sensitive parameter in the WM-RL family.

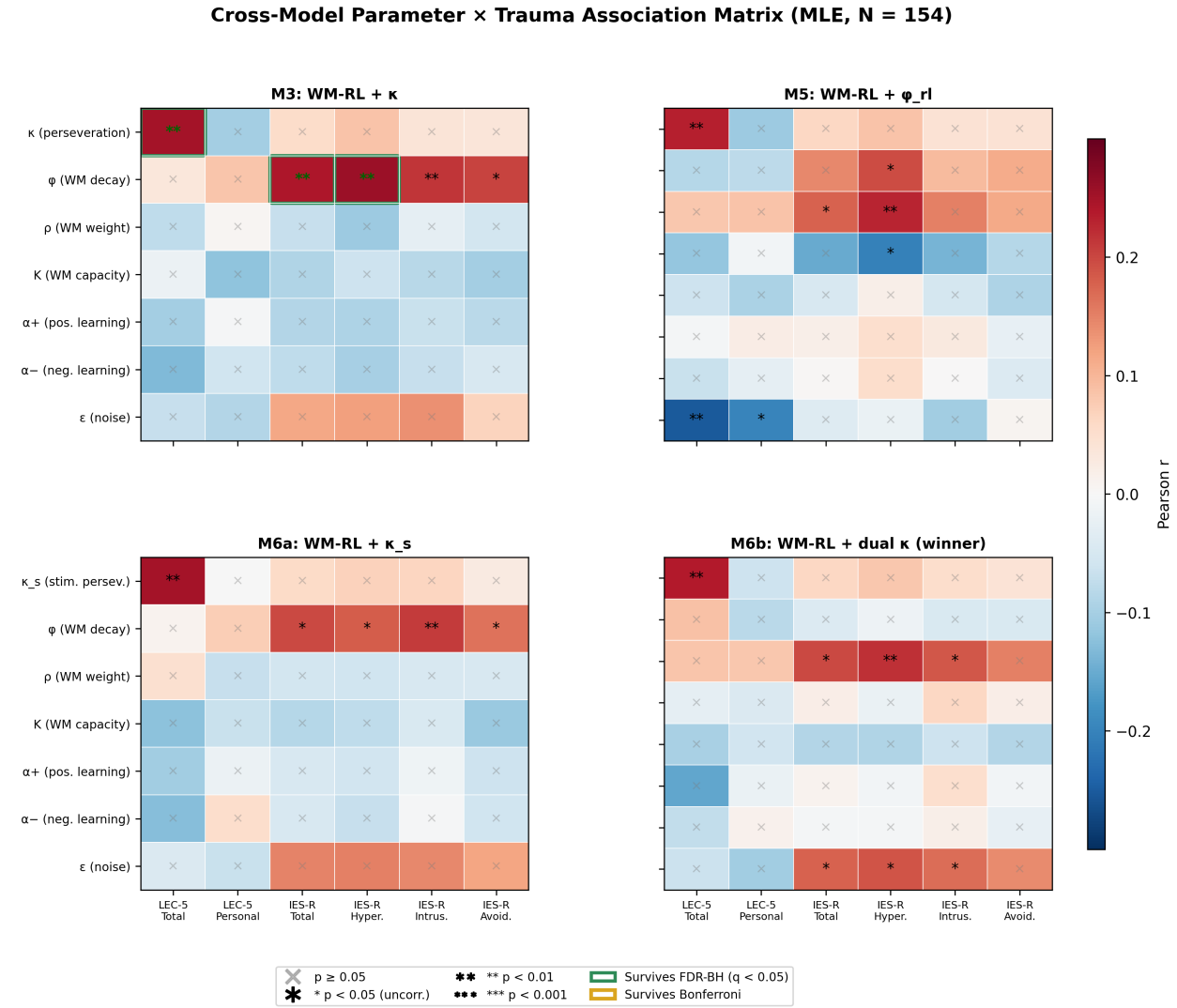


Figure 10: Cross-model parameter × trauma association matrix. Each panel shows one WM-RL model variant (M3, M5, M6a, M6b). See text for legend details.

6.10 Appendix J: Model Parameters

Table 10: Free parameters for all seven computational models. M4 is the only joint choice+RT model; its AIC is not comparable to choice-only models M1–M6b.

Table 10			
Model	Free Parameters	k	Type
M1: Q-Learning	alpha, epsilon	2	Choice only
M2: WM-RL	alpha, phi, rho, K, epsilon	5	Choice only
M3: WM-RL+kappa	alpha, phi, rho, K, kappa, epsilon	6	Choice only
M5: WM-RL+phi_rl	alpha, phi, rho, K, kappa, phi_RL, epsilon	7	Choice only
M6a: WM-RL+kappa_s	alpha, phi, rho, K, kappa_s, epsilon	6	Choice only
M6b: WM-RL+dual	alpha, phi, rho, K, kappa_total, kappa_share, epsilon	7	Choice only
M4: RLWM-LBA	alpha, phi, rho, K, kappa, v_scale, A, delta, t0	9	Choice+RT

6.11 Appendix K: Exclusion Criteria

All v4.0 analyses used a canonical cohort implemented as `config.get_analysis_cohort()`. A participant was included if **all three** of the following gates were satisfied:

1. **Task completeness.** The participant contributed at least 400 trials to the main task (blocks 3–23) and participated in at least 8 main-task blocks. The trial-count threshold (`MIN_TRIALS_THRESHOLD = 400` in `config.py`) corresponds to approximately 50% of the expected 807–1077 trials across 21 blocks and ensures sufficient data for reliable parameter estimation.
2. **Performance check.** Overall accuracy ≥ 0.40 AND mean accuracy across the last 5 main-task blocks ≥ 0.50 . With three response options, chance is $1/3 \approx 0.333$. The overall-accuracy gate rules out participants at or below chance; the late-block gate additionally confirms that the participant *learned* rather than performing at chance throughout the session. A participant who selected the correct response deterministically in a small minority of blocks but was otherwise random would satisfy the overall gate but fail the late-block gate.
3. **Scale completeness.** Non-missing values on every required survey column: LEC-5 total events, IES-R total, IES-R intrusion, IES-R avoidance, IES-R hyperarousal. Participants with any missing survey data were excluded from Level-2 regression analyses; this was also applied upstream to the MCMC fit for the hierarchical Bayesian pipeline so that individual-level posteriors and group-level regressions are estimated on an identical sample.

Participants with known duplicate or corrupted data (`MANUAL_EXCLUSIONS` in `config.py`) were always excluded regardless of gates 1–3.